

Executive Summary

Simultaneous Group-Envelope Bounds for Γ -Robust Multiple-Choice Knapsack Problems

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Many resource-allocation decisions choose exactly one option per group. Protecting their shared constraint with a Bertsimas–Sim uncertainty budget produces a robust multiple-choice knapsack problem. The classical threshold reduction is polynomial, but many distinct deviations can make complete LP certification expensive. Büsing, Gersing, and Koster (2023) report that LP solves consume 33.1% of runtime on average among solved instances and 93.17% at one million items, where the root LP alone averages 1,062 seconds. Their objective-uncertainty setting differs from ours, but it identifies the bounded-threshold LP family as a material bottleneck.

After dualizing a fixed-threshold MCKP, all threshold-dependent baselines cancel. Between consecutive group deviations, the remaining envelope is the maximum of one constant and one common-slope line. Prefix and suffix maxima with range accumulation evaluate one multiplier over all B thresholds in $O(B + K \log(B + 1))$ time and $O(B + K)$ working storage, rather than $\Theta(BK)$ direct score evaluations, for K options.

Weak duality makes the interval minimax value a valid upper bound on every feasible fixed-threshold LP; strong LP duality gives equality on feasible singletons. An epigraph-dual mapping proves exact-minimax dominance over the bounded-threshold group-clique LP. Convex bracketing returns a valid bound with an explicit minimization-error certificate, and best-first splitting certifies the maximum LP value over the complete threshold family.

The comparison is component-matched: both certificates use the same adaptive search, fixed-threshold solver, candidate rule, tolerance, and timing policy; only the interval upper bound differs. A common-trace experiment also evaluates both oracles on identical intervals. This isolates the new component rather than comparing with DnC+, a full Java/Gurobi solver that combines filtering, estimators, cuts, incumbent operations, and early termination for objective uncertainty.

Across 60 primary instances, the compressed certificate reaches tolerance throughout, wins all 60 instance-level median comparisons, and is 2.37 times faster in geometric mean than the sparse clique LP (95% interval 2.27–2.47). Speedup rises from 1.69 at 360 groups to 3.24 at 1,440 groups. All 36 robustness cases reach tolerance. On 9 published robust-knapsack coefficient sets, it wins all 9 comparisons with a 2.75-fold advantage; this is an out-of-generator coefficient test, not replication of the source model. Algebraic, epigraph-LP, and exhaustive-scan checks find maximum error 6.58×10^{-8} and no certificate violation.

The deliverable is a faster complete threshold-disjunctive LP certificate for use at the root or in an outer robust search. A separate exact-integer audit confirms objective agreement and valid gap accounting, while showing that integer subproblem work can erase the oracle advantage and that compact SCIP remains preferable on many-breakpoints instances. The claim is therefore simultaneous all-threshold evaluation for exactly-one groups, not a new uncertainty model or universal integer-solver superiority.